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Effectiveness Analysis and Clinical Application Potential Exploration of Combined Detection of PAX1/JAM3 Gene Methylation in Early Diagnosis of Cervical Precancerous Lesions

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Abstract

Background: The early and accurate diagnosis of cervical precancerous lesions is essential for mitigating the risk of cervical cancer. Given the limitations of traditional screening methods, such as inadequate sensitivity and low specificity, this study aims to

assess the diagnostic performance of combined PAX1/JAM3 gene methylation testing for cervical intraepithelial neoplasia (CIN) and to explore its potential in triage management. Methylation testing serves as a tool for investigating the epigenetic alterations associated with diseases like cancer, thereby providing molecular biological evidence for diagnosis and prognosis. **Methods:** This study included 431 cervical exfoliated cell samples, which consisted of CIN1, CIN2/3, cervical cancer (CC), and control groups. The methylation levels of the PAX1/JAM3 genes were assessed using quantitative methylation-specific PCR. The diagnostic efficacy of these genes for high-grade lesions (CIN2+) was analyzed through receiver operating characteristic (ROC) curve analysis and was compared with HPV testing and cytological examination. Additionally, the diagnostic performance of PAX1/SOX1 methylation was preliminarily evaluated in a sub-cohort of 100 cases. Further evaluation was conducted on the triage referral rate and the impact of clinical intervention using this biomarker in HPV-positive patients. **Results:** The area under the curve (AUC) for the PAX1/JAM3 methylation test in diagnosing CIN2+ was 0.89 (95% CI: 0.85–0.92), with a sensitivity of 83.1% and a specificity of 88.8%. In the sub-cohort, the PAX1/SOX1 combination also demonstrated high diagnostic potential for CIN3+ (AUC = 0.91), serving as a validation

of methylation testing efficacy. In non-HPV16/18 positive patients, the sensitivity and specificity of methylation testing for CIN2+ were superior to those of cytology. Among patients undergoing colposcopy, 265 cases had results of $\leq$ ASCUS, of which 51 were methylation positive, resulting in a colposcopy referral rate of 19.3%.

Conclusion: PAX1/JAM3 gene methylation testing demonstrates strong diagnostic efficacy for cervical precancerous lesions and holds significant potential in triage diagnosis. The PAX1/SOX1 panel also shows promise as a complementary biomarker.

Keywords: Cervical intraepithelial neoplasia; DNA methylation; PAX1; JAM3; Molecular Diagnostic Techniques.

1 Introduction

Cervical cancer, classified as a gynecological tumor, poses a significant threat to women's health worldwide. According to statistics, it has become the fourth most common cancer, with an estimated 661,000 new cases and 348,000 deaths globally by 2022¹. Notably, China ranks first in the world for new cervical cancer cases, with 150,700 reported, and second for deaths, totaling 55,700^{1,2}. Therefore, early detection and treatment of cervical cancer are crucial for women's health. The progression of cervical cancer typically begins with an infection by the human papillomavirus (HPV). If this infection persists, it may lead to the development of

precancerous lesions, which, if not treated promptly, can progress to cervical cancer many years later^{3,4}. With the widespread use of HPV-related vaccines, early and regular cervical screening remains essential for preventing cervical cancer^{5,6}.

In line with the World Health Organization's "Global Strategy to Accelerate the Elimination of Cervical Cancer," China is actively promoting nationwide efforts for the prevention and control of cervical cancer⁷. However, significant disparities exist in cervical cancer screening across different regions of the country. Both ThinPrep cytology test (TCT) and HPV testing are widely utilized methods in the cervical cancer screening process; however, they have limitations, including the subjectivity of cytological examination and the low specificity of HPV DNA testing⁸. The combined use of both methods can enhance the accuracy and efficiency of screening for cervical precancerous lesions and cervical cancer⁹⁻¹¹. Based on the screened patients, a definitive diagnosis can be established using pathology as the gold standard. Following the initial screening, a referral for colposcopy is essential¹². To decrease the referral rate for colposcopy among HR-HPV positive women, there is an urgent need for an effective, non-invasive triage strategy that operates independently of the subjective interpretation of pathologists. To advance the elimination of cervical cancer in our

country, there is an urgent need for a stable and cost-effective detection method^{13,14}. Research has indicated that the inactivation of tumor suppressor genes and the activation of oncogenes significantly contribute to carcinogenesis through genetic and epigenetic alterations¹⁵. Among these alterations, methylation, a form of epigenetic modification, emerges as a promising method for early tumor screening. It has been successfully utilized in the early detection of various cancers, including lung, colorectal, and bladder cancers¹⁶. The experimental samples required for methylation testing, such as blood, urine, and tissue, are readily obtainable in clinical settings. Furthermore, the testing is relatively objective and low-cost, making it an appropriate choice for the early screening and prognostic assessment of cervical cancer.

In 2024, the "Chinese Journal of Cancer Prevention and Treatment" first proposed a clinical pathway for cervical cancer screening based on expert consensus. Both Junctional Adhesion Molecule 3 (JAM3) and Paired Box Gene 1 (PAX1) are recognized as being associated with cervical cancer^{17,18}. However, there is a lack of research on the application of combined methylation detection of JAM3 and PAX1 genes for diagnosing cervical lesions in the Xinjiang region. The triage capability of JAM3/PAX1 methylation for cervical lesions and its accuracy compared to Liquid-Based Cytology (LBC)

remain to be elucidated. This study aims to evaluate the diagnostic performance of combined PAX1/JAM3 gene methylation testing for cervical intraepithelial neoplasia (CIN) and to explore its potential application in triage management.

Beyond JAM3, other genes involved in cellular differentiation and tumor suppression have been explored as potential methylation biomarkers for cervical cancer. The SOX1 (SRY-box transcription factor 1) gene, a key regulator of Wnt signaling pathways frequently silenced by promoter methylation in various malignancies, has also been implicated in cervical carcinogenesis¹⁹⁻²¹. Given the promising diagnostic performance of PAX1-based panels and the established biological role of SOX1, we sought to preliminarily evaluate the hitherto less-studied PAX1/SOX1 methylation combination and compare its diagnostic efficacy against the more established PAX1/JAM3 benchmark for detecting high-grade cervical lesions.

2 Subjects and Methods

2.1 Study subject

A total of 1,021 patients who visited the Affiliated Tumor Hospital of Xinjiang Medical University from September 2024 to February 2025 were included as study subjects. Based on the established inclusion and exclusion criteria, 431 patients with abnormal HPV-DNA test results or cytological examination findings necessitating colposcopy

were ultimately selected for the study. All participants voluntarily engaged in the research and provided signed informed consent forms. The inclusion criteria were as follows: 1) Case group: Cervical tissue samples with a confirmed pathological diagnosis that met the diagnostic criteria for squamous intraepithelial lesions (grades I, II, III) and cervical cancer; 2) Control group: Histologically confirmed normal cervical tissues or negative biopsies from patients who underwent colposcopy but were confirmed to have no neoplastic lesions; 3) Strict adherence to the cervical cancer screening protocol, with all participants undergoing TCT, hrHPV testing, and colposcopy; 4) No history of anti-human papillomavirus (HPV) treatment within the preceding six months. The exclusion criteria included: 1) A history of known reproductive tract malignancies or other tumors; 2) Patients who were uncooperative during the examination and had incomplete data; 3) Pregnant or lactating women; 4) A history of cervical cancer vaccination.

2.2 Methods

2.2.1 Clinical Information Collection

The patient's basic information includes: 1) General information such as outpatient number, name, age, and medical history; 2) TCT results, which are categorized into Atypical Squamous Cells of Undetermined Significance (ASC-US), negative for intraepithelial

lesions or malignancy (NILM), low-grade squamous intraepithelial lesions (LSIL), and high-grade squamous intraepithelial lesions (HSIL); 3) hr-HPV DNA results, classified as HPV types 16/18, non-16/18 high-risk HPV, and negative (indicating no hr-HPV infection); 4) Cervical histopathology results, categorized as Cervical Intraepithelial Neoplasia Grade I, II, III (CIN I, II, III) and Cervical Cancer (CC).

2.2.2 TCT, HPV, histopathology, and DNA methylation testing

The procedure for cervical sample collection and testing is as follows: Each participant provided a single cervical sample. This was collected using a specialized cervical brush, which was rotated 5-10 times clockwise at the external cervical os and within the cervical canal. The brush head was then placed into a ThinPrep® PreservCyt® vial for liquid-based cytology (LBC) preparation. For cytological examination, a thin-layer cell smear was prepared from the LBC vial. The slide was then Papanicolaou-stained and evaluated by a professional pathologist according to the 2014 Bethesda Reporting System (TBS) to issue a diagnosis (e.g., NILM, ASC-US, LSIL, etc.)²². For HPV DNA testing, an aliquot from the same LBC vial was used for genotyping. HPV-specific DNA fragments were amplified using the Human Papillomavirus 21 Genotype Nucleic Acid Detection Kit via polymerase chain reaction (PCR)¹¹. This kit

identifies 14 high-risk types (HPV16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 66, 68), among others. For histopathological confirmation, patients with abnormal cytology and/or positive HPV results underwent colposcopy. Based on evaluations by two experienced pathologists, biopsy samples were classified as follows: Control group: Histologically confirmed normal cervical tissues or negative biopsies. Case groups: Cervical Intraepithelial Neoplasia Grade I, II, III (CIN I, II, III) and Cervical Cancer (CC).

DNA Methylation Analysis: Genomic DNA was extracted from equal volume samples using manufacturer-certified kits: Blood/Cell/Tissue DNA Kit (Tiangen DP304) for PAX1/JAM3 detection and Nucleic Acid Purification Kit (Cap BSPC-D-M) for PAX1/SOX1 detection. Distinct extraction kits were employed to strictly adhere to the validated protocols specific to each downstream methylation detection assay, ensuring optimal compatibility. DNA concentration and purity were verified by spectrophotometry (NanoDrop™) with OD260/280 ratios maintained at 1.6-2.1. A minimum of 200 ng (PAX1/JAM3) or 500 ng (PAX1/SOX1) DNA underwent bisulfite conversion using EZ-96 DNA Methylation-Lightning™ MagPrep (Zymo D5046) for PAX1/JAM3 or Bisulfite Conversion Kit (Cap D-D-M) for PAX1/SOX1, converting unmethylated cytosine to uracil. Conversion efficiency was verified

through amplification of internal reference genes (ACTB for PAX1/SOX1; GAPDH for PAX1/JAM3), with $\geq 95\%$ conversion rate validated per kit specifications.

Technical controls followed the specific manufacturer protocols for each assay: internal controls used ACTB gene (valid Ct 15-35) for SOX1/PAX1 and GAPDH gene (valid Ct < 35) for PAX1/JAM3. Run controls included negative controls (nuclease-free water, no amplification) and manufacturer-supplied positive controls (fully methylated DNA standards) with specified Ct ranges: 22-28 for all genes in SOX1/PAX1 detection and 25-31 for target genes and GAPDH in PAX1/JAM3 detection.

Methylation positivity was defined as follows: for SOX1/PAX1, valid samples (ACTB Ct 15-35) required SOX1 Ct ≤ 38.6 and/or PAX1 Ct ≤ 38.0 ; for PAX1/JAM3, valid samples (GAPDH Ct < 35) required ΔCt (PAX1) ≤ 6.6 and/or ΔCt (JAM3) ≤ 10 . Methylation levels were assessed by ΔCt values, with lower ΔCt indicating higher methylation.

2.2.3 Sample Selection and Grouping Allocation

This study included a total of 431 patients with histopathologically confirmed cervical lesions and normal controls. The cohort consisted of 81 cases in the control group, 161 cases of CIN1, 77 cases of CIN2, 96 cases of CIN3, and 16 cases of cervical cancer (CC),

all of whom underwent PAX1/JAM3 gene methylation testing. To further investigate the diagnostic potential of PAX1 in conjunction with other genes, 100 samples were randomly selected from the total cohort (CIN1: n=37; CIN2: n=18; CIN3+: n=26; normal controls: n=19) for additional PAX1/SOX1 methylation testing. The selection criteria were based on the principle of stratified random sampling to ensure that the proportions of each lesion grade (CIN1, CIN2, CIN3+) were consistent with those in the original cohort, thereby minimizing selection bias.

2.3 Statistical analysis

All statistical analyses were conducted using SPSS version 26.0 (IBM Corporation, Armonk, NY, USA) and GraphPad Prism version 10.1.2 (GraphPad Software, San Diego, CA, USA), with statistical significance set at $p < 0.05$. Normality was initially assessed using the Shapiro-Wilk test. Data that followed a normal distribution are expressed as mean $\pm$ standard deviation ($\bar{x} \pm s$). For continuous variables that did not meet the assumption of normal distribution, non-parametric tests (Mann-Whitney U test or Kruskal-Wallis H test) were employed for between-group comparisons, with the data presented as median and interquartile range (IQR) (Median [IQR]). For continuous variables between groups, one-way analysis of variance (ANOVA) was employed after verifying normality and

homogeneity of variance. Measurement data were analyzed using either the t-test or the Wilcoxon rank-sum test. Enumeration data were expressed as counts (%), and differences in categorical data between groups were statistically analyzed using Pearson's chi-square test or Fisher's exact probability method. The diagnostic efficacy of each methylation marker was evaluated using the receiver operating characteristic curve (ROC curve). To assess the combined diagnostic performance of PAX1 and JAM3, a binary logistic regression model was utilized to create a composite predictor. For each diagnostic endpoint, the disease status was defined as the dependent variable, while the ΔCt values of PAX1 and JAM3 were entered as independent continuous predictors. The model was fitted using maximum likelihood estimation ($\text{Logit}(P) = \beta_0 + \beta_1\Delta\text{Ct_PAX1} + \beta_2\Delta\text{Ct_JAM3}$), and the resulting predicted probability (P) for each sample was used as the combined indicator for the subsequent ROC analysis. The area under the curve (AUC), sensitivity, specificity, and their 95% confidence intervals (CIs) were calculated.

3 Results

3.1 Comparison of General Data Among Groups

Table 1 presents the general characteristics of the study subjects. A total of 431 female patients were included in this study, comprising

81 in the normal group, 161 in the CIN1 group, 77 in the CIN2 group, 96 in the CIN3 group, and 16 in the CC group. According to the age screening criteria established by the World Health Organization, there were 175 participants under 45 years of age and 256 participants aged 45 and above. The age distribution for each group was as follows: normal group 47.00 years (IQR: 35.50-53.50); CIN1 group 49.00 years (IQR: 38.00-56.00); CIN2 group 51.00 years (IQR: 40.75-57.00); CIN3 group 46.00 years (IQR: 39.00-56.75). None of the groups exhibited a normal distribution in age. The Kruskal-Wallis H test indicated statistically significant differences in age among the five histological groups (Control, CIN1, CIN2, CIN3, and CC) ($P = 0.032$). Regarding the methylation positive rates, the rates for the normal, CIN1, CIN2, CIN3, and CC groups were 8.6%, 1.9%, 77.9%, 95.8%, and 100% respectively, with statistically significant differences observed between the groups ($P < 0.001$).

3.2 Comparison of methylation positivity rates

For each group, the methylation positivity rate was defined as the percentage of samples that tested positive and was calculated by $(\text{number of positive samples} / \text{total samples}) \times 100\%$. Table 2 presents the methylation positivity rates of high-risk HPV (hrHPV) genotypes alongside the results of cytological assessments. Significant differences existed in the methylation positivity rates

among the HPV Negative group (41.2%), the HPV 16/18 Positive group (54.7%), and the HPV Non-16/18 Positive group (28.8%). Similarly, notable differences in methylation positivity rates were found among the NILM, ASCUS, LSIL, and HSIL groups, with rates of 17.5%, 22.2%, 58.2%, and 88.5%, respectively. Figure 2 illustrates the distribution of ΔCt values for PAX1/JAM3 across the control group, CIN1, CIN2, and CIN3+ groups. The observed differences in ΔCt values among these groups indicate variations in methylation levels. The more pronounced decrease in ΔCt values in the CIN2+ and higher groups suggests that JAM3/PAX1 hypermethylation may serve as a potential biomarker for high-grade lesions.

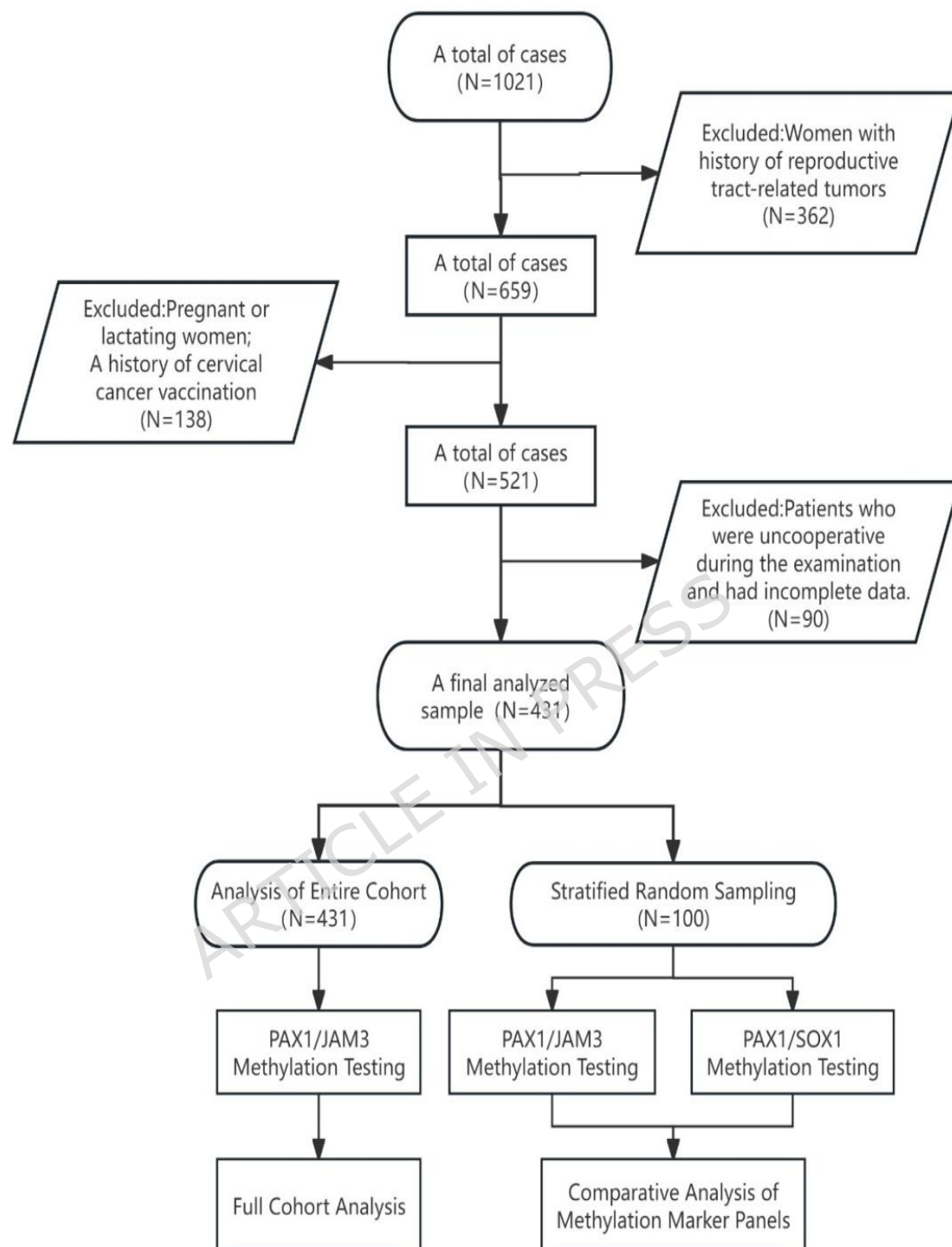


Figure 1 Flowchart of inclusion criteria for current analysis

Table 1 Overview of general data characteristics (N = 431).

Characteristics	Result	<i>N</i>	Proportion (%)	Methylation positivity rate (%)	<i>p-Value</i>
Histopathology					
	Control	81	18.8	8.6	<0.001
	CIN1	161	37.4	1.9	
	CIN2	77	17.9	77.9	
	CIN3	96	22.3	95.8	
	Cervical Cancer	16	3.7	100	
High-risk HPV (hr-HPV) test					
	Negative	34	7.9	41.2	<0.001
	16/18 positive	192	44.5	54.7	
	Non-16/18 positive	205	47.6	28.8	
Age (years)					
	<45	175	40.6	37.1	0.147
	≥45	256	59.4	44.1	
Cytology test					
	NILM	80	18.6	17.5	<0.001
	ASCUS	185	42.9	22.2	
	LSIL	79	18.3	58.2	
	HSIL/ASC-H	87	20.2	88.5	

3.3 Diagnostic Performance of Methylation Testing for CIN1+, CIN2+, and CIN3+

Figure 3 illustrates the ROC curve analysis regarding the diagnostic performance of PAX1/JAM3 methylation in the detection of cervical intraepithelial neoplasia (CIN). For CIN2+, the AUC for PAX1 methylation was 0.89 (95% CI: 0.86-0.93), and for JAM3 methylation, it was 0.83 (95% CI: 0.79-0.87). For CIN3+, PAX1 methylation demonstrated an AUC of 0.90 (95% CI: 0.85-0.94), while JAM3 methylation exhibited an AUC of 0.92 (95% CI: 0.88-0.95). When combined, the PAX1/JAM3 methylation test for CIN2+ revealed an

AUC of 0.89 (95% CI: 0.85–0.92), with a sensitivity of 83.1% (95% CI: 77.1–87.7 %) and a specificity of 88.8% (95% CI: 84.3–92.2 %). For CIN3+, the AUC was 0.93 (95% CI: 0.89–0.96), with a sensitivity of 85.7% (95% CI: 78.1–91.0%) and a specificity of 87.8% (95% CI: 83.7–90.9%). Table 3 presents the AUC, sensitivity, and specificity of PAX1/JAM3 and PAX1/SOX1 for CIN3+ across different cohorts. Figure 4 displays the receiver operating characteristic (ROC) curves for the diagnostic efficacy of PAX1/JAM3 and PAX1/SOX1 methylation testing in detecting CIN3+. In a cohort of 100 cases, the combined detection of PAX1/SOX1 for CIN3+ yielded an AUC of 0.91 (95% CI: 0.83–0.99), with a sensitivity of 76.9% (95% CI: 56.4–91.0%) and a specificity of 94.6% (95% CI: 86.7–98.5%). In comparison to the performance of PAX1/JAM3 in the overall cohort (AUC=0.93, 95% CI: 0.89–0.96; sensitivity 85.7%, specificity 87.8%) and in the sub-cohort (AUC=0.97, 95% CI: 0.95–1.00); sensitivity 100.0% (95% CI: 87.1–100.0%), specificity 85.1% (95% CI: 75.3–91.5%), the PAX1/SOX1 methylation test also exhibited excellent diagnostic performance for CIN3+ (AUC = 0.91).

3.4 Comparison of Diagnostic Efficiency Between Cytology and Methylation Testing in Triage of HPV Non-16/18 Positive Patients

Current clinical guidelines recommend immediate colposcopy for HPV16/18 positive women due to their high immediate risk. However,

for non-16/18 hrHPV positive women, triage with cytology is standard practice to determine the need for referral. Therefore, we specifically evaluated the performance of methylation testing against cytology in this sub-population to assess its potential as a superior triage tool. The comprehensive comparison of triage methods presented in Table 4 reveals distinct performance profiles for cytology and methylation testing. Cytology demonstrated high sensitivity for all endpoints (93.8% for CIN2+ and 100% for CIN3+) but consistently low specificity (27.9% and 24.7%, respectively). In contrast, PAX1/JAM3 methylation testing showed a more balanced performance, particularly for clinically relevant endpoints. For CIN2+ cases, methylation testing provided 69.2% sensitivity and 86.4% specificity (Youden's index = 0.557), while for CIN3+ cases, it achieved 80.6% sensitivity and 78.2% specificity (Youden's index = 0.588).

3.5 Reducing unnecessary colposcopy referrals for HPV-positive patients through methylation testing

Specifically, we compared standard cytological triage (referral $\geq$ ASC-US) with methylation triage patterns. As shown in Table 5, the methylation test showed excellent triage ability in the HPV non-16/18 positive group. Compared with cytology, methylation testing reduced colposcopy referral rates from 70.5% to 15.1%, significantly

reducing unnecessary procedures. Crucially, this reduction was accompanied by no decrease in sensitivity for CIN2+ (100% detection) and a significant improvement in specificity (96.9% vs 33.6%). This data strongly supports the use of PAX1/JAM3 methylation as a preferred triage tool for non-16/18 hrHPV-positive women.

Table 2 Methylation positivity rates in various hrHPV genotypes and cytological test results comparison (N = 431).

	Control (n=81)	CIN1 (n=161)	CIN2 (n=77)	CIN3 (n=96)	CC (n=16)	Total	P
hrHPV genotype							
HPV							
Negative (n=34)	0/10(0)	1/11(9.1%)	11/11(100%)	2/2(100%)	0(0)	14/34(41.2%)	<0.001
16/18 positive (n=192)	3/14(21.4%)	1/67(1.5%)	25/32(78.1%)	63/66(95.5%)	13/13(100%)	105/192(54.7%)	
Non-16/18 positive (n=205)	4/57(7.0%)	1/83(1.2%)	24/34(70.6%)	27/28(96.4%)	3/3(100%)	59/205(28.8%)	
Cytology							
NILM(n=80)	2/40(5.0%)	0/22(0)	7/12(58.3%)	4/5(80.0%)	1/1(100.0%)	14/80(17.5%)	<0.001
ASCUS(n=185)	4/40(10.0%)	0/100(0)	19/26(73.1%)	16/17(94.1%)	2/2(100.0%)	41/185(22.2%)	
LSIL(n=79)	1/1(100.0%)	2/31(6.5%)	18/20(90.0%)	25/27(92.6%)	0(0)	46/79(58.2%)	
HSIL (n=87)	0(0)	1/8(12.5%)	16/19(84.2%)	47/47(100.0%)	13/13(100.0%)	77/87(88.5%)	

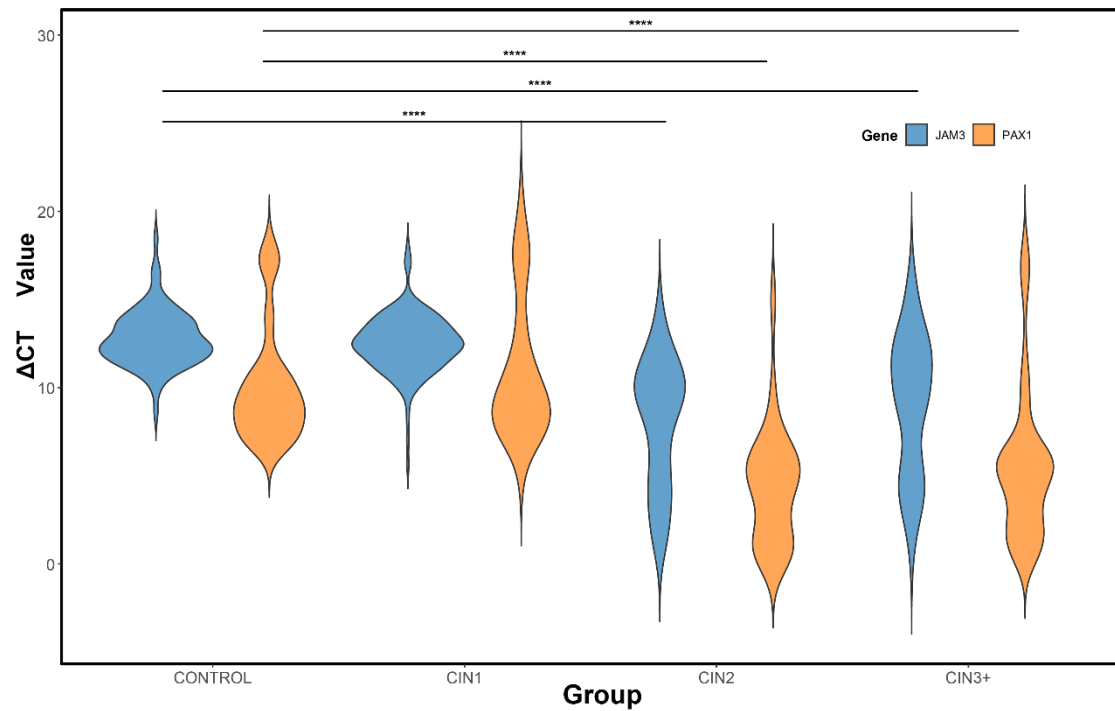


Figure 2 Methylation levels of JAM3 and PAX1 promoters across cervical lesion groups.

Violin plots show ΔCT distribution. Lower ΔCT values indicate higher methylation. Groups: Control, CIN1, CIN2, CIN3+. **** $P \leq 0.0001$.

Table 3 AUC, sensitivity and specificity of PAX1/JAM3 and PAX1/SOX1 for CIN3+ across different cohorts.

Gene Combination	Cohort Type	N	AUC (95%CI)	Sensitivity (95% CI)	Specificity (95% CI)
PAX1/JAM3	Overall cohort	431	0.93 [0.89-0.96]	85.7% [78.1-91.0%]	87.8% [83.7-90.9%]
PAX1/JAM3	Sub-cohort	100	0.97 [0.95-1.00]	100.0% [87.1-100.0%]	85.1% [75.3-91.5%]
PAX1/SOX1	Sub-cohort	100	0.91 [0.83-0.99]	76.9% [56.4-91.0%]	94.6% [86.7-98.5%]

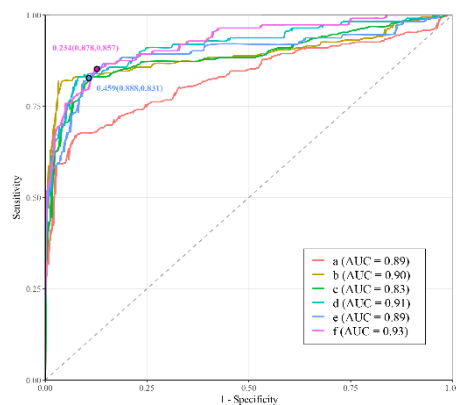


Figure 3 Receiver operating characteristic (ROC) curves of PAX1/JAM3 methylation

detection for diagnosing cervical intraepithelial neoplasia (CIN).

Receiver operating characteristic (ROC) curves analysis of **(a)** PAX1 methylation for CIN2+ (AUC = 0.89), **(b)** PAX1 methylation for CIN3+ (AUC = 0.90), **(c)** JAM3 methylation for CIN2+ (AUC = 0.83), **(d)** JAM3 methylation for CIN3+ (AUC = 0.92), **(e)** combined PAX1/JAM3 methylation for CIN2+ (AUC = 0.89, Sensitivity = 83.1%, Specificity = 88.8%), **(f)** combined PAX1/JAM3 methylation for CIN3+ (AUC = 0.93, Sensitivity = 85.7%, Specificity = 87.8%). The dashed diagonal line represents the reference line of no discriminatory ability (AUC = 0.5). The solid black dot on the combined curve indicates the optimal cut-off point with the corresponding sensitivity and specificity values. AUC, area under the curve.

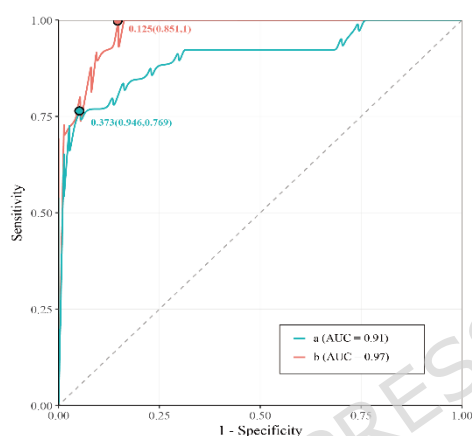


Figure 4 Receiver Operating Characteristic (ROC) Curve Analysis of PAX1/JAM3 and PAX1/SOX1 Methylation for the Diagnosis of Cervical Intraepithelial Neoplasia Grade 3 or Higher (CIN3+).

a: ROC curve illustrating the combined detection of PAX1/SOX1 gene methylation for the diagnosis of CIN3+ in the sub-cohort; (AUC = 0.91, Sensitivity = 76.9%, Specificity = 94.6%); b: ROC curve illustrating the combined detection of PAX1/JAM3 gene methylation for the diagnosis of CIN3+ in the sub-cohort (AUC = 0.97, Sensitivity = 100.0%, Specificity = 85.1%).

Table 4 Comparison diagnostic performance of cytology and methylation testing for triage of HPV Non-16/18+ positive cases (N = 205).

Group	Triage method	Sensitivity (%)	Specificity (%)	False Negativity (%)	False positivity (%)	PPV (%)	NPV (%)	Accuracy (%)	Youden's index
CIN2+	Cytology	93.8	27.9	6.2	72.1	37.7	90.7	48.8	0.217
	Methylation	69.2	86.4	30.8	13.6	70.3	85.8	81.0	0.557
CIN3+	Cytology	100	24.7	0	75.3	19.1	100.0	36.1	0.247
	Methylation	80.6	78.2	19.4	21.8	39.7	95.8	78.5	0.588

*Note: The optimal cut-off values for methylation testing were determined by maximizing Youden's index. The primary clinically relevant endpoints are CIN2+ and CIN3+. PPV: Positive Predictive Value; NPV: Negative Predictive Value.

Table 5 Comparison of referral rates and diagnostic performance between Cytology and

Methylation triage strategies in HPV-positive patients (N = 247).

HPV Status	Triage Strategy	Referral Rate (%)	Sensitivity for CIN2+ (%)	Specificity for CIN2+ (%)
HPV Non-16/18+ (N=146)	Cytology ($\geq$ ASC-US)	70.5 (103/146)	100.0 (18/18)	33.6 (43/128)
	Methylation (Positive)	15.1 (22/146)	100.0 (18/18)	96.9 (124/128)
HPV 16/18+ (N=101)	Cytology ($\geq$ ASC-US)	63.4 (64/101)	55.6 (15/27)*	33.8 (25/74)
	Methylation (Positive)	28.7 (29/101)	77.1 (27/35)	97.3 (72/74)

*Note: Sensitivity calculation based on CIN2+ endpoints. Cytology showed notably low sensitivity in the HPV 16/18+ group, reinforcing guidelines for immediate referral in this high-risk population.

4 Discussion

This study demonstrates that PAX1/JAM3 methylation analysis serves as a powerful biomarker for high-grade cervical lesions, showing a significant positive correlation between methylation levels and histological severity^{23,24}. Importantly, our findings confirm that PAX1/JAM3 methylation testing exhibits significantly higher discriminatory power for advanced lesions (CIN2+ and CIN3+) compared to low-grade ones (CIN1+), a pattern that aligns with the biological premise that these epigenetic changes are a hallmark of progressive disease. The AUC values we obtained for high-grade lesions are comparable to, and in some cases superior to, those reported for other methylation panels in similar populations^{24,25}. As the severity of the lesions increases, the degree of methylation tends to become more consistent, thereby enhancing the detection signal strength; in contrast, the heterogeneous methylation status in early lesions reduces the stability of the marker.

This dynamic change results in methylation detection achieving higher sensitivity (88.9%) and specificity (95.9%) for advanced lesions. The above results indicate a positive correlation between the degree of methylation and the severity of cervical lesions, providing a molecular basis for the precise triage of high-risk populations.

This study emphasizes that, for non-16/18 hrHPV positive cases, methylation testing demonstrates significant specificity advantages in cytology for CIN2+ and CIN3+ cases, potentially reducing unnecessary colposcopies. Current clinical guidelines advocate for hrHPV testing as the primary screening method for cervical cancer, primarily due to its high disease detection rate and standardized testing procedures. However, this strategy has significant limitations, as the majority of hrHPV-positive cases represent transient infections that are not directly associated with cervical lesions. In response, the current management pathway proposes a stratified approach: individuals who test positive for HPV types 16/18 are directly referred for colposcopy evaluation, while those positive for the other 12 high-risk types require combined cytology testing (with referral if the result is ASCUS or higher). The negative population, on the other hand, only needs to be rechecked every three years. Methylation testing shows significant potential for precise triage within this framework. By quantifying the methylation

levels of genes such as PAX1 and JAM3, it can effectively differentiate between progressive lesions and low-risk infections. This approach reduces unnecessary colposcopy referrals while maintaining sensitivity for the detection of high-grade lesions^{26,27}. The introduction of this triage model is anticipated to optimize the balance between resource allocation and overtreatment in the current screening system.

Regarding the criteria for colposcopy referrals, the guidelines recommend that a risk of CIN3 exceeding 4% necessitates such referrals²⁸⁻³¹. However, the lack of systematic risk screening and assessment methods among many primary care physicians in our country has led to significant variability in colposcopy referral standards. From a clinical perspective, the most significant finding is the potential of methylation testing to drastically reduce unnecessary colposcopy referrals. Our model, which prioritizes referral based on methylation positivity, would lower the referral rate to approximately 20%. For patients with negative methylation results, the misdiagnosis rate associated with colposcopy is low. Nevertheless, effectively identifying high-risk patients who progress to high-grade lesions without subjecting them to overtreatment remains a challenge. The Xinjiang region remains a high-incidence area for cervical cancer³². Given the diagnostic discrepancies among

gynecologists in China, which may lead to over-referral, unnecessary patient anxiety, and increased medical costs, our research findings advocate for the use of methylation testing as a potential tool for informed triage in this patient population. In this study, the methylation detection of PAX1/JAM3 genes was found to be more effective for severe lesions (CIN2+/CIN3+) but less effective for milder lesions such as CIN1 and below. Lesions classified as CIN1 and below are typically characterized by atypia. Although low-grade lesions exhibit cytological atypia, they often follow a self-limiting course due to their high regression rate. In contrast, high-grade lesions demonstrate significantly reduced regression potential due to the persistent activation of methylation-driven genes³³. This study confirms that methylation testing, by quantifying the cumulative extent of epigenetic modifications, can identify the propensity for malignant transformation caused by persistent HPV infection at an earlier stage³⁴. For the ASC-US/LSIL group with ambiguous cytological diagnoses, the high specificity of this test can reduce unnecessary colposcopy referrals by approximately 42%, thereby optimizing the allocation of medical resources and alleviating the psychological burden on patients. In summary, while the challenge of early CIN1+ diagnosis lies in its natural tendency to resolve, methylation testing offers a valuable reference for early prevention.

This study strictly adhered to the gold standard of pathological diagnosis, with all enrolled samples possessing complete histological verification results, thereby ensuring data reliability through standardized quality control procedures. In comparison to previous studies, this research systematically evaluated the dynamic diagnostic efficacy of PAX1/JAM3 dual-gene methylation across the continuous progression stages of cervical lesions and proposed its application threshold as a triage tool for hrHPV-positive populations. This provides quantifiable molecular evidence for optimizing the existing "primary screening - triage - referral" three-tier prevention and control system. However, the conclusions of this study are limited by its single-center retrospective design and insufficient sample size, particularly in the CC subgroup, which comprises only 16 cases. Furthermore, while we explored the potential of combining SOX1 with PAX1/JAM3 in a preliminary subset analysis (n=100), the sample size was too small to draw definitive conclusions. This limitation may lead to statistical power bias. Additionally, the current analysis concentrated on the phenotypic association between methylation levels and lesion grading, without exploring the underlying driving mechanisms. In this study, functional experiments were not used to verify the regulatory relationship between PAX1 and JAM3/SOX1, and in vitro experiments are needed

to clarify the molecular interaction mechanism. Future research should validate the universality of triage efficacy through multicenter prospective cohort studies, integrate multi-omics technologies, such as whole-genome methylation sequencing, to investigate the molecular regulatory networks of key genes, and conduct health economic evaluations to clarify its cost-effectiveness advantages.

5 Conclusion

The PAX1/JAM3 methylation test exhibits significant diagnostic value in high-grade cervical precancerous lesions (CIN2+ and CIN3+). Its high negative predictive value and low false positive rate effectively reduce unnecessary colposcopy referrals, establishing it as a crucial tool in triage diagnosis. In addition, preliminary analysis in a sub-cohort suggests that PAX1/SOX1 methylation may also serve as a promising complementary biomarker, but further validation is needed in a larger population.

Abbreviations

ASCUS	Atypical Squamous Cells of Undetermined Significance
CC	cervical cancer
CIs	confidence intervals
CIN	cervical intraepithelial neoplasia
HSIL	high-grade cervical cytological lesion

IQR	interquartile range
JAM3	Junctional Adhesion Molecule 3
LBC	Liquid-Based Cytology
LSIL	low-grade cervical cytological lesion
NILM	negative for intraepithelial lesions or malignancy
PAX1	Paired Box Gene 1
ROC	receiver operating characteristic
SOX1	SRY-box transcription factor 1
TCT	ThinPrep cytology test

Ethics approval and consent to participate

The Ethics Committee of the Affiliated Cancer Hospital of Xinjiang Medical University conducted ethical approval for this study with K-2024021 as the ethical approval number.

Consent for publication

Written informed consent was obtained from all participants.

Availability of data and materials

All data generated or analysed during this study are included in this published article.

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Competing interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Authors' contributions

GF conceived the study, designed the methodology, and obtained funding. YXL collects clinical samples, performs methylation experiments, and verifies data quality. YK statistical analysis included ROC curve modeling and comparative test. KWN explained the clinical implications, designed the triage strategy, and contributed to the data visualization. XXH drafted the manuscript and coordinated revisions. MXM oversaw the project, critically reviewed the manuscript, and obtained ethical approval. All the authors read the final manuscript, critically revised it for intellectual content, and approved it.

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